Marios Papachristou

Assistant Professor of Information Systems
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Personal Information

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Nationality	Greek, US Permanent Resident (Green Card Holder)

Research

My research develops models of resilience in AI-enabled socio-technical systems, focusing on how network structure, algorithmic agents, and information dynamics jointly determine systemic resilience of socio-technical systems.

Methods

Network Science, AI agents, Privacy, Computational Social Science

Employment

W.P. Carey School of Business, Arizona State University *2025 –*
Tenure-Track Assistant Professor, Department of Information Systems
- *Affiliated Member*, Center for AI and Data Analytics (AIDA)
- *Graduate Faculty Member*, Data Science, Analytics and Engineering

Microsoft Research *Summer 2023*
Office of Applied Research, *with Chin-Chia Hsu and Longqi Yang*

Twitter *Summer 2022*
User Modeling Research Group, *with Rishab Goel, Frank Portman, Matt Miller*

Education

Cornell University *2020 – 2025*
Ph.D. in Computer Science, Minor: *Applied Mathematics*
- *Doctoral Advisor:* Jon Kleinberg
- *Thesis Title:* From Contagion to Stability: Insights into Network Dynamics, Resilience and Stability

M.S. in Computer Science *2020 – 2022*

National Technical University of Athens *2015 – 2020*
Diploma in Electrical and Computer Engineering
- *Advisor:* Dimitris Fotakis

Honors & Awards

- **ACM Doctoral Dissertation Award Nominee** *2025*
  Nominated to the ACM by Cornell's CS department as the best Ph.D. thesis in CS awarded in 2025.
- **ACM/AAAI SIGAI Doctoral Dissertation Award Nominee** *2025*
  Nominated to the ACM SIGAI by Cornell's CS department as the best Ph.D. thesis in AI awarded in 2025.
- **Sigma Xi** *2025*
  Elected member based on scientific contributions.
- **Cornell ORIE Young Researchers Workshop** *2024*

- **Alexander A. Onassis Scholarship** *2023*
  Three-year fellowship for Greek Ph.D. students. Acceptance rate: 16%.
- **Bodossaki Foundation Fellowship** *2023*
  Runner-up.
- **LinkedIn Ph.D. Fellowship** *2022*
  One-year fully funded fellowship by LinkedIn. Acceptance rate 14%.
- **A.G. Leventis Scholarship** *2022*
  Award for graduate students of Greek origin. Acceptance rate 12.5%.
- **Gerondelis Scholarship** *2022*
  One-time award for graduate students of Greek descent.
- **Chateaubriand Fellowship** *2022*
  Fellowship granted by the French Embassy in the US for studies in France (declined for personal reasons). Acceptance rate 33%.
- **Cornell Fellowship** *2020*
  One-year fully funded fellowship by the CS department based on "exceptional preparation and promise".
- **Thomaidion Award** *2019*
  Awarded by NTUA for publication during undergraduate studies.
- **ESEC/FSE 2019 ACM Student Research Competition Finalist** *2019*
- **CanSat International Space Engineering Competition** *2019*
  4th out of 93 teams. Organized by the American Astronautical Society (AAS).
- **2nd prize "be finnovative 2.0" Start-up Accelerator by National Bank of Greece** *2018*
  Cashierless payments start-up.
- **1st prize at "Crowdhackathon Fintech #2" by the National Bank of Greece** *2017*
  Cashierless payments start-up.
- **IEEEXtreme 11.0 Programming Competition** *2017*
  Top 1% worldwide in IEEE's algorithms competition.
- **Touramanoglu Scholarship** *2015*
  Highest University Entrance Exams score in the Imittos region.
- **"The Great Moment of Education" Scholarship by Eurobank SA** *2015*
  Highest University Entrance Exams score in the Imittos region.

Journal Papers

Google Scholar Profile
$^{\alpha\beta}$= alphabetical order
*= equal contribution
$^{\circ}$= corresponding author

[J1] **Network Formation and Dynamics among Multi-LLMs**
Marios Papachristou, Yuan Yuan
PNAS Nexus, 2025
[paper] [preprint] [code]
- *Taught at UC Berkeley, Stanford, UIUC and others*
- *Featured by TechXplore, and SingularityHub*

[J2] **Leveraging Large Language Models for Collective Decision-Making**
Marios Papachristou, Longqi Yang, Chin-Chia Hsu
Proceedings of the ACM on Human Computer Interaction (presented at ACM CSCW '25), 2025
[paper]
- *Invited talk at the LLMs as Research Tools Workshop at ACM CHI 2024*
- *Featured by the ACM*

[J3] **`volesti`: A C++ library for sampling and volume computation on convex bodies**
$^{\alpha\beta}$Apostolos Chalkis*, Vissarion Fisikopoulos*, ***Marios Papachristou****, Elias Tsigaridas*

Journal of Open Source Software, 2025
[paper] [code]

[J4] **Differentially Private Distributed Estimation and Learning**
Marios Papachristou, Amin Rahimian
IISE Transactions (formerly IIE Transactions), 2025
and the University of Pittsburgh Swanson School of Engineering
[paper] [code] [talk]
- *Featured at the July 2025 issue of the ISE Magazine (10% of articles published in IISE Transactions)*
- *Featured by the University of Pittsburgh Swanson School of Engineering*

[J5] **Truncated Log-concave Sampling for Convex Bodies with Reflective Hamiltonian Monte Carlo**
$^{\alpha\beta}$Apostolos Chalkis, Vissarion Fisikopoulos, ***Marios Papachristou***$^{\circ}$, Elias Tsigaridas
ACM Transactions on Mathematical Software (TOMS), 2023
[paper] [code]
- *Featured at ODBMS.org*

[J6] **Sublinear Domination and Core–periphery Networks**
Marios Papachristou
Scientific Reports, 2021
[paper] [code]

[J7] **Commons-based peer production and digital fabrication: The case of a RepRap-based, Lego-built 3D printing-milling machine**
Vasilis Kostakis*, and ***Marios Papachristou****
Telematics and Informatics, 2014
[paper]

Top-tier CS Conference Proceedings

[C1] **Opinion Dynamics with Multiple Adversaries**
Akhil Jalan*, ***Marios Papachristou****$^{\circ}$
Forthcoming at WWW '26: ACM Web Conference (WWW), 2026
[paper]
- *Acceptance rate: 20%*

[C2] **Dynamic Interventions for Networked Contagions**
Marios Papachristou, Sid Banerjee, Jon Kleinberg
WWW '23: ACM Web Conference (WWW), 2023
[paper] [code]
- *Acceptance rate: 17%*

[C3] **Core-periphery Models for Hypergraphs**
Marios Papachristou, Jon Kleinberg
KDD '22: ACM Intl. Conf. on Knowledge Discovery and Data Mining (KDD), 2022
[paper] [code] [data]
- *Acceptance rate: 15%*
- *Taught at tutorials at KDD and CIKM*

[C4] **Allocating Stimulus Checks in Times of Crisis**
Marios Papachristou, Jon Kleinberg
WWW '22: ACM Web Conference (WWW), 2022
[paper] [code]
- *Acceptance rate: 17%*
- *Featured by the Federal Reserve Bank of Minneapolis*
- *Taught at UC Berkeley*

[C5] **Software Clusterings with Vector Semantics and the Call Graph**
Marios Papachristou
ESEC/FSE '19: ACM Joint Meeting on European Software Engineering Conference

and Symposium on the Foundations of Software Engineering (ESEC/FSE), 2019
[paper] [code] [data]
- *Acceptance rate: 21%*
- *ESEC/FSE Student Research Competition Finalist Paper*

Revisions & Working Papers

[WP1] **Structural Measures of Resilience for Supply Chains**
Marios Papachristou, Amin Rahimian, Arash Azadegan
Major revision, Production and Operations Management, 2024
[preprint] [code] [slides]

[WP2] **Echoes of Disagreement: Measuring Disparity in Social Consensus**
Marios Papachristou, Jon Kleinberg
Major revision, INFORMS Journal of Computing, 2024
[preprint] [code]

[WP3] **Optimal Resource Allocation for Remediating Networked Contagions**
Marios Papachristou, Sid Banerjee, Jon Kleinberg
Under preparation for submission, 2025
[preprint] [slides]

[WP4] **Differentially Private Distributed Inference**
Marios Papachristou, Amin Rahimian
Under preparation for submission, 2025
Preliminary version presented at the Theory and Practice of Differential Privacy Workshop (TPDP)
[preprint] [talk]

Refereed CS Workshops

[W1] **GLINKX: A Scalable Unified Framework for Homophilous and Heterophilous Graphs**
Marios Papachristou, Rishab Goel, Frank Portman, Matt Miller, Rong Jin
NeurIPS workshop on Graph Learning Frontiers (GLFrontriers), 2022
[paper] [poster]

Tutorials

[T1] **Differential Privacy for Strategic Information Sharing and Learning: Foundations, Mechanisms, and Applications**
Amin Rahimian, Juba Ziani, ***Marios Papachristou***, Yuxin Liu
Conference on Web and Internet Economics (WINE), 2025

Works in Progress

[WIP1] **Impact of Massive AI-generated Content on Recommender Systems**
with Joonhwae Park, and Panos Adamopoulos

[WIP2] **The Effects of Generative AI on Consumer Behavior**
with Panos Adamopoulos, and Nikhil Garg

[WIP3] **Pivot, Commit, or Wait? Evaluating non-Committal Learning Strategies in Digital Ventures**
with Julian Lehmann, and Gerald Kane

[WIP4] **Ambient Language Models; Learning to Talk; Learning to Talk by Looking at Noise**
with Giannis Daras, Alkis Kalavasis, Costis Daskalakis, and Antonio Torralba

[WIP5] **Optimal Resolution of a Data Sharing Trilemma: Statistical Power, Sample Complexity and Privacy Budget**
with Yuxin Liu, Amin Rahimian, and Arash Azadegan

[WIP6] **The Price of Misreporting: Measuring the Effect of Adversarial Behavior in Online Platforms**
with Akhil Jalan

Software

volesti `github.com/GeomScale/volesti`
Open-source C++ statistical software for sampling and volume approximation from convex bodies.

Other Experience

GeomScale Open-source Organization *May 2020 – March 2020*
Sampling from truncated log-concave densities, and convex optimization, working on the `volesti` open-source package *with Dr. Apostolos Chalkis, Dr. Vissarion Fisikopoulos, and Dr. Elias Tsigaridas*

Cashierless Payments Start-up *2017 – 2018*
Part of National Bank of Greece's start-up accelerator

Athens University of Economics and Business *2018 – 2020*
Department of Management Science and Technology, *with Prof. Diomidis Spinellis*

Google Summer of Code 2020 (GeomScale) *June 2020 – August 2020*
Sampling from high-dimensional truncated log-concave densities, *with Dr. Apostolos Chalkis, Dr. Vissarion Fisikopoulos, and Dr. Elias Tsigaridas*
`[code]` `[talk]`

Google Summer of Code 2018 (GFOSS-OTA) *April 2018–September 2018*
Automated codification of Greek Legislation, *with Prof. Diomidis Spinellis*
`[code]` `[data]` `[talk]`

Researcher – P2P Lab *2013 – 2014*

Teaching & Mentorship Experience

- The Structure of Information Networks (PhD-level, Cornell) *Spring & Fall 2023*
- Discrete Mathematics (NTUA) *Spring 2017*
- Programming Techniques (NTUA) *Spring 2016*
- Introduction to Computer Programming (NTUA) *Fall 2016, Fall 2017*

- Student-applicant Support Program at Cornell CIS *2022*
- Google Summer of Code Mentor *2019–2022*
- Volunteer lessons for the National Olympiad of Informatics (PDP) *2019*

Mentees

- Joonhwae Park (Ph.D. student at Emory)
- Yuxin Liu (Ph.D. student at the University of Pittsburgh)
- Jasmine Li (Undergraduate Cornell CS)
- Ioannis Iakovidis (Google Summer of Code 2022; now Ph.D. student at Johns Hopkins)
- Zhang Zhuyan (Google Summer of Code 2022; now Technology analyst at Bank of America)

Talks & Lectures

† = scheduled
* = presented by co-author

- ***Resilience in Networks: Frameworks for Resource Allocation and Simulation***
  - Archimedes, Athena RC *April 2025*
  - TOC4Fairness Seminar, Simons Institute *March 2025*
  - Cornell University, Theory Seminar *December 2024*
  - Arizona State University, W.P. Carey School of Business *November 2024*
  - Purdue University, Daniels School of Business *November 2024*

- MIT, Research Group Meeting *June 2024*

- ***Network Formation and Dynamics among Multi-LLMs***
  - Arizona State University, School of Mathematical and Natural Sciences, AI Seminar *November 2025*
  - INFORMS Annual Meeting *October 2025*
  - Symposium on Statistical Challenges on Electronic Commerce Research (SCECR) *June 2025*
  - Arizona State University, Machine Learning Day *April 2025*
  - NSF Conference on Network Science and Economics *April 2025*
  - INFORMS Annual Meeting *October 2024*
  - IBM Research, Exploratory Math Sciences Council Seminar *April 2024*
  - UIUC, Guest Lecture at the Network Analysis Course *April 2024*
  - Cornell University, AIPP Group *March 2024*
  - Learning on Graphs (LoG) NYC Meetup *February 2024*
- ***Information Aggregation and Distributed Learning in Privacy-critical Environments***
  - CMU, Tepper School of Business OM Seminar⋆ *September 2024*
  - TOC4Fairness Online Seminar Series⋆ *March 2024*
  - Information: Theory and Applications (ITA) Workshop⋆ *February 2024*
  - Rutgers University Business School, Seminar⋆ *November 2023*
- ***Leveraging Large Language Models for Collective Decision-Making***
  - LLMs as Research Tools: Applications and Evaluations in HCI Data Work Workshop, ACM Conference on Human-Computer Interaction⋆ *May 2024*
  - Microsoft Research, Invited Applied Research Talks *August 2023*
- ***Structural Measures of Resilience for Supply Chains***
  - NSF Conference on Network Science and Economics *April 2024*
  - Columbia University, Student Theory Seminar *November 2023*
  - Cornell University, Theory Tea *November 2023*
  - INFORMS Annual Meeting⋆ *October 2023*
  - Cornell University, LinkedIn Campus Visit *May 2023*
- ***Resource Allocation in a Financial Contagion Environment***
  - University of Chicago, Theory Seminar *October 2023*
  - Indiana University, Center for Complex Networks and Systems Research *September 2023*
  - Cornell University, CS 6850 Guest Lecture *April 2023*
  - Cornell University, Theory Seminar *November 2022*
- ***Dynamic Interventions for Networked Contagions***
  - INFORMS Annual Meeting *October 2023*
  - ACM Web Conference *April 2023*
  - ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization *October 2022*
- ***Core-periphery Models for Hypergraphs***
  - ACM Conference on Knowledge Discovery and Data Mining *August 2022*
- ***GLINKX: A Scalable Unified Framework for Homophilous and Heterophilous Graphs***
  - NeurIPS Workshop on Graph Learning Frontiers *December 2022*
  - Twitter, Machine Learning Seminar *August 2022*
- ***Allocating Stimulus Checks in Times of Crisis***
  - ACM Web Conference *April 2022*
  - ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization *October 2021*
- ***Sampling from Truncated High Dimensional Logconcave Densities***
  - PyData Global *December 2020*
- ***Software Clusterings with Vector Semantics and the Call Graph***

– ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering *August 2019*

Departmental And Professional Service

Department and School Outreach

- Department of Information Systems Brown Bag Seminar Organizer [link]
- W. P. Carey School of Business, PhD Lunch and Learn, November 2025, Keynote Speaker [link]

Program Commitee

- ACM Conference on Fairness Accountability and Transparency (FAccT)
- ACM Conference on Knowledge Discovery and Data Mining (KDD)
- IEEE International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (TPS)

Ad-hoc Reviewer

- MIS Quarterly
- Management Science
- Mathematics of Operation Research
- Journal of Artificial Intelligence Research
- Naval Research Logistics
- International Conference on Computational Social Science (IC2S2)
- ACM Web Conference (WWW)
- Innovations in Theoretical Computer Science (ITCS)
- ACM Conference on Human Factors in Computing (CHI)
- European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD)
- Machine Learning
- NeurIPS 2022 Workshop on Graph Learning Frontiers (GLFrontiers)
- NeurIPS 2021 Workshop on Human and Machine Decisions
- Journal of Open Source Software

Session Chair

- INFORMS Annual Meeting *2025*
- NSF Confernce on Network Science and Economics *2024*
- ACM Web Conference (WWW) *2022*

References

Jon Kleinberg (kleinberg@cornell.edu)
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Amin Rahimian (rahimian@pitt.edu)
Deptartment of Industrial Engineering, University of Pittsburgh

Yuan Yuan (yuyuan@ucdavis.edu)
OpenAI and Graduate School of Management, UC Davis

Longqi Yang (longqi.yang@microsoft.com)
Office of Applied Research, Microsoft Research

Chin-Chia Hsu (chinchia.k.hsu@gmail.com)
Google DeepMind

Yong-Yeol Ahn (yyahn@virginia.edu)

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